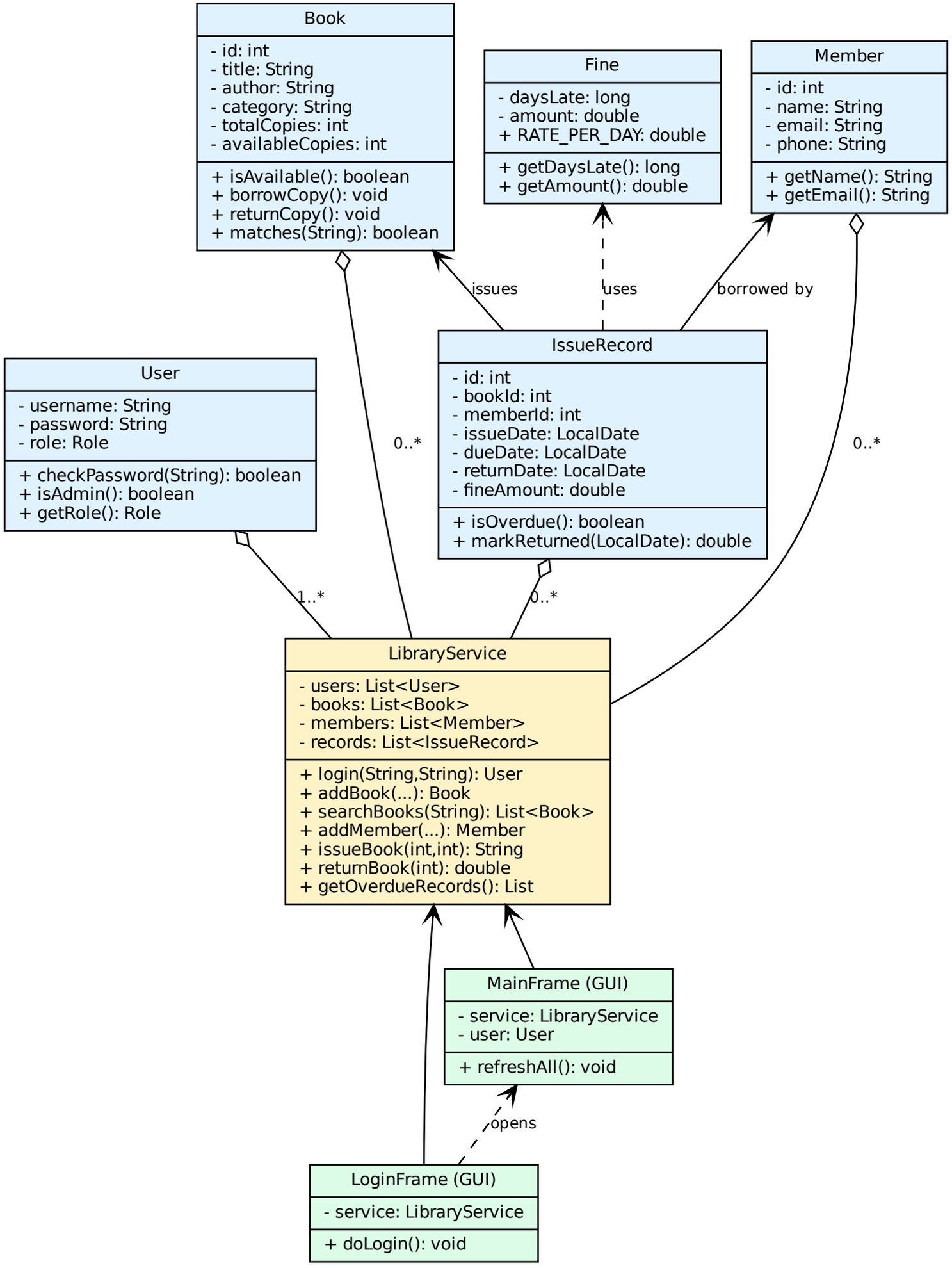
Book
- id: int
- title: String
- author: String
- category: String
- totalCopies: int
- availableCopies: int
+ isAvailable(): boolean
+ borrowCopy(): void
+ returnCopy(): void
+ matches(String): boolean
Fine
- daysLate: long
- amount: double
+ RATE_PER_DAY: double
+ getDaysLate(): long
+ getAmount(): double
Member
- id: int
- name: String
- email: String
- phone: String
+ getName(): String
+ getEmail(): String
issues
uses
borrowed by
IssueRecord
- id: int
- bookId: int
- memberId: int
- issueDate: LocalDate
- dueDate: LocalDate
- returnDate: LocalDate
- fineAmount: double
+ isOverdue(): boolean
+ markReturned(LocalDate): double
User
- username: String
- password: String
- role: Role
+ checkPassword(String): boolean
+ isAdmin(): boolean
+ getRole(): Role
0..*
0..*
1..*
0..*
LibraryService
- users: List<User>
- books: List<Book>
- members: List<Member>
- records: List<IssueRecord>
+ login(String,String): User
+ addBook(...): Book
+ searchBooks(String): List<Book>
+ addMember(...): Member
+ issueBook(int,int): String
+ returnBook(int): double
+ getOverdueRecords(): List
MainFrame (GUI)
- service: LibraryService
- user: User
+ refreshAll(): void
opens
LoginFrame (GUI)
- service: LibraryService
+ doLogin(): void